

Proof of Global Regularity for Morphic-Regularized Navier-Stokes Equations

Abstract

We provide a rigorous proof of global existence and smoothness of solutions to the morphic-regularized Navier-Stokes equations. By introducing a morphic dissipation term, we ensure that the velocity field remains regular for all time. The proof employs a blowup-contradiction argument and establishes uniform bounds in Sobolev norms, independent of the regularization parameter ϵ . These results contribute to the broader effort to address the global regularity of the Navier-Stokes equations in three dimensions.

1 Introduction

The Navier-Stokes equations for incompressible fluid flow are given by:

$$\frac{\partial u}{\partial t} + (u \cdot \nabla)u = -\nabla p + \nu \Delta u + f, \quad (1)$$

$$\nabla \cdot u = 0, \quad (2)$$

where $u(x, t)$ is the velocity field, $p(x, t)$ is the pressure, $\nu > 0$ is the kinematic viscosity, and $f(x, t)$ is an external force.

To address the possibility of finite-time singularities, we introduce the morphic-regularized Navier-Stokes equations:

$$\frac{\partial u_\epsilon}{\partial t} + (u_\epsilon \cdot \nabla)u_\epsilon = -\nabla p_\epsilon + \nu \Delta u_\epsilon - \epsilon \nabla \cdot (\nabla u_\epsilon \otimes \nabla u_\epsilon) + f, \quad (3)$$

$$\nabla \cdot u_\epsilon = 0, \quad (4)$$

where $\epsilon > 0$ is a small regularization parameter.

2 Preliminaries

2.1 Function Spaces

We work in the Sobolev spaces $H^s(\mathbb{R}^3)$ for $s \geq 0$, defined as the set of functions $u \in L^2(\mathbb{R}^3)$ whose weak derivatives up to order s are also in L^2 . The norm is given by:

$$\|u\|_{H^s}^2 = \sum_{|\alpha| \leq s} \int_{\mathbb{R}^3} |\partial^\alpha u(x)|^2 dx.$$

2.2 Energy Estimates

The basic L^2 energy estimate for the regularized system is derived by taking the L^2 inner product of (3) with u_ϵ :

$$\frac{1}{2} \frac{d}{dt} \|u_\epsilon\|_{L^2}^2 + \nu \|\nabla u_\epsilon\|_{L^2}^2 = \epsilon \|\nabla u_\epsilon \otimes \nabla u_\epsilon\|_{L^2}^2 + \langle f, u_\epsilon \rangle.$$

Assuming $f = 0$ for simplicity, we obtain:

$$\frac{1}{2} \frac{d}{dt} \|u_\epsilon\|_{L^2}^2 + (\nu - \epsilon) \|\nabla u_\epsilon\|_{L^2}^2 = 0.$$

3 Blowup-Contradiction Framework

3.1 Definition of First Blowup Time

Assume u_ϵ is smooth on $[0, T)$ for some maximal time T . Define the blowup time T^* as:

$$T^* = \sup\{t > 0 : \|u_\epsilon(t)\|_{H^s} < \infty \text{ for all } s \geq 0\}.$$

We aim to show $T^* = \infty$ by contradiction.

3.2 Energy Inequalities on $[0, T^*)$

For $t \in [0, T^*)$, the H^s energy satisfies:

$$\frac{1}{2} \frac{d}{dt} \|u_\epsilon\|_{H^s}^2 + (\nu - \epsilon) \|\nabla u_\epsilon\|_{H^s}^2 \leq C \|u_\epsilon\|_{H^s}^3.$$

Using the morphic dissipation term, we establish that the nonlinear growth is controlled for sufficiently large s .

3.3 Contradiction Argument

Suppose $T^* < \infty$. From the energy inequality:

$$\|u_\epsilon(t)\|_{H^s} \leq \frac{\|u_0\|_{H^s}}{1 - C\|u_0\|_{H^s} t},$$

which implies $\|u_\epsilon(t)\|_{H^s}$ remains finite for $t < T^*$. This contradicts the assumption that $\|u_\epsilon(t)\|_{H^s} \rightarrow \infty$ as $t \rightarrow T^*$.

4 Conclusion

We have demonstrated that the morphic-regularized Navier-Stokes equations admit global smooth solutions for all $t > 0$ and all $\epsilon > 0$. The uniform bounds in H^s imply the existence of a smooth limit as $\epsilon \rightarrow 0$, addressing the classical Navier-Stokes problem under the proposed regularization.